Dark Matter Module: the physics behind

Contents

1 Dark Matter

- **Target topics:** Dark Matter (galaxy rotation curves, clusters).
- **Learning objectives:** Analyze multiple lines of evidence for dark matter; derive galaxy rotational velocity curve from mechanics, Newton gravitation law and Kepler law; investigate galaxies and clusters composition; find the cluster mass from virial theorem reviewing kinetic energy and gravitational potential; connect classical Physics to Cosmology.
- **Physics program link:** Mechanics,Kepler's law, Newton's law of gravitation, centripetal acceleration, circular motion, centripetal force,gravitational force, energy and momentum conservation, gravitational potential energy, kinetic energy.

1.1 Galaxy rotational velocity curve

- **Objectives:** Apply the law of gravitation to derive the expected orbital velocity in a galaxy. Compare the theoretical velocity with observational data from galaxy datasets.Introduce the dark matter hypothesis as a potential solution to the discrepancy.
- **Dataset variables of galaxies:** [10]
  - Rad (radius)
  - V_{obs} (observed velocity)
  - errV (velocity error)
  - V_{gas} (gas velocity)
  - V_{disk} (stellar disk velocity)
  - V_{bul} (bulge velocity)

  - $\mathrm{SB}_{\mathrm{disk}}$ (surface brightness disk)
  - $\mathrm{SB}_{\mathrm{bul}}$ (surface brightness bulge)

Classical Physics derivation:

- We model a small mass m (orbiting body) in a stable circular orbit around a large central mass M (e.g., galactic center) at a radius r.
- The condition for stable orbit requires the Gravitational Force (F_G) to equal the Centripetal Force (F_c).

Gravitational and Centripetal Forces

- **Gravitational Force (F_G):**

$$F_G = \frac{GMm}{r^2} \tag{1}$$

- **Centripetal Force (F_c):**

$$F_c = \frac{mv^2}{r} \tag{2}$$

- **Equating the Forces ($F_G = F_c$):**

$$\frac{GMm}{r^2} = \frac{mv^2}{r} \tag{3}$$

Deriving the rotational velocity (v)

- **Cancellation of m:** The mass of the orbiting body (m) cancels out, showing v is independent of m.

$$\frac{GM}{r^2} = \frac{v^2}{r} \tag{4}$$

- **Isolating v^2:** Multiply by r to isolate v^2.

$$v^2 = \frac{GM}{r} \tag{5}$$

- **Final Rotational Velocity (v):** Take the square root.

$$v = \sqrt{\frac{GM}{r}} \tag{6}$$

- **Physical Meaning:** This Keplerian velocity profile dictates that $v \propto 1/\sqrt{r}$, meaning orbital speed decreases as distance from the central mass increases.

Connection to Kepler's third law

- **Orbital Period (T):** Period is the circumference divided by velocity.

$$T = \frac{2\pi r}{v} \tag{7}$$

- **Kepler's relation :** Square T and substitute $v^2 = GM/r$.

$$T^2 = \frac{4\pi^2 r^2}{v^2} = \frac{4\pi^2 r^2}{GM/r} \implies T^2 = \frac{4\pi^2}{GM} r^3 \tag{8}$$

- **Physical Meaning:** This confirms Kepler's Third Law ($T^2 \propto r^3$), demonstrating that the proportionality derived from the force balance is fully consistent with the relationship between orbital period and radius.

Velocity and mass

- **Baryonic velocity (from data):** Incorporating the Mass-to-Light ratio (Υ) and conserving the signs for counter-rotating components [9]:

$$V_{\text{bar}}^2(r) = V_{\text{gas}}|V_{\text{gas}}| + \Upsilon\left[V_{\text{disk}}|V_{\text{disk}}| + V_{\text{bul}}|V_{\text{bul}}|\right] \tag{9}$$

- **Baryonic mass:**

$$M_{\text{bar}}(r) = \frac{r\ \max(V_{\text{bar}}^2(r), 0)}{G} \tag{10}$$

- **Total observed mass:**

$$M_{\text{obs}}(r) = \frac{rV_{\text{obs}}^2(r)}{G} \tag{11}$$

- **Dark matter velocity squared**: Derived from the difference between the observed total velocity and the baryonic prediction:

$$V_{\text{DM}}^2(r) = V_{\text{obs}}^2(r) - V_{\text{bar}}^2(r) \tag{12}$$

- **Dark matter density (from data):**

$$\rho_{\text{DM}}(r) = \frac{1}{4\pi G r^2}\frac{d}{dr}\left(r \cdot V_{\text{DM}}^2(r)\right) \tag{13}$$

NFW Dark Matter Profile

- **NFW profile for dark matter [12, 11]:**

$$\rho_{\text{NFW}}(r) = \frac{\rho_s}{(r/r_s)(1 + r/r_s)^2} \tag{14}$$

  where r_s is the scale radius and ρ_s is the characteristic density, $\alpha = 1, \beta = 3, \gamma = 1$.

- **Global Fit parameters:** The structural parameters ρ_s, r_s, and the mass-to-light ratio Υ are calculated via a global χ^2 minimization on the entire rotation curve dataset to provide a robust estimation of the halo.

- **NFW dark matter mass:**

$$M_{\rm NFW}(r) = 4\pi \int \rho_{\rm NFW} r^2 dr = 4\pi\rho_s r_s^3 \left[\ln\left(1+\frac{r}{r_s}\right) - \frac{r/r_s}{1+r/r_s}\right] \tag{15}$$

- **Velocity dark matter:**

$$V_{\rm DM}(r) = \sqrt{\frac{GM_{\rm NFW}(r)}{r}} \tag{16}$$

- **Total simulated curve:**

$$V_{\rm sim}(r,f) = \sqrt{\frac{GM_{\rm bar}(r) + f\cdot M_{\rm NFW}(r)}{r}} \quad \text{with } f\in[0,2] \tag{17}$$

Data visualization and fit

- **Plot:**
  - **X-axis:** Rad (radius, data)
  - **Y-axis:**
    - $V_{\rm bar}$ (red)
    - $V_{\rm obs}$ (data, blue) $\pm$errV
    - $V_{\rm sim}$ (green)

- **Fit χ^2/dof (chi-squared):**

$$\chi^2/\text{dof} = \frac{1}{N-P}\sum_{i=1}^{N}\left(\frac{V_{\rm obs}(r_i) - V_{\rm sim}(r_i)}{\text{errV}(r_i)}\right)^2 \tag{18}$$

  where N is the number of data points and P is the number of free parameters ($P=3$: ρ_s, r_s, Υ).

Global χ^2 Minimization

The parameters r_s, ρ_s, and the mass-to-light ratio Υ are determined simultaneously by minimizing the χ^2 over the entire rotation curve:

$$\chi^2 = \sum_{i=1}^{N}\left(\frac{V_{\rm obs}(r_i) - V_{\rm tot,sim}(r_i)}{\text{errV}(r_i)}\right)^2 \tag{19}$$

where $V_{\rm tot,sim}(r) = \sqrt{\max(V_{\rm bar}^2(r) + V_{\rm DM}^2(r), 0)}$.

1.2 Velocity distribution in galaxy clusters

- **Objectives:** Help students infer the existence of dark matter in galaxy clusters using classical mechanics and virial theorem. Students will apply the virial theorem to estimate the total mass of a galaxy cluster from the kinetic and potential energy of its galaxies.
- **Dataset Coma cluster (SDSS)** : objid (galaxy ID), ra (right ascension, α), dec (declination, δ), modelmag (apparent magnitude in r band, m_r), modelmagerr, extinction, redshift (z), zErr (redshift error) [13, 1].
- **Dataset Abell Clusters:** Cluster (name), ID (galaxy ID), RAdeg (right ascension in degrees), DEdeg (declination in degrees), RV (radial velocity in km/s), e_RV (velocity error in km/s), q_RV (quality flag), bmag (apparent magnitude in B band). [14]

1. **Individual Galaxy Velocity:** Assuming non-relativistic speeds for low-redshift clusters, the radial velocity of each galaxy is derived directly from its observed redshift data:

$$v_i = c\, z_i \tag{20}$$

2. **Number of observed galaxies:** N
3. **Mean Cluster Velocity:** The systemic velocity of the cluster is calculated as the average of the individual radial velocities:

$$\bar{v} = \frac{1}{N}\sum_{i=1}^{N} v_i \tag{21}$$

4. **Observed Velocity Dispersion:** The 1-D velocity dispersion is the standard deviation of the galaxies' velocities relative to the systemic mean, providing a measure of the cluster's dynamical state:

$$\sigma_{\text{obs}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(v_i - \bar{v})^2} \tag{22}$$

5. **Comoving Distance:** To determine physical scales, we integrate over the expansion history of the Universe (Eq. 15 [7]):

$$\chi(z) = \frac{c}{H_0}\int_0^z \frac{dz'}{\sqrt{\Omega_m(1+z')^3 + (1-\Omega_m)}} \tag{23}$$

6. **Angular Diameter Distance:** This converts the comoving distance into a metric that relates apparent angular size to true physical size (Eq. 18 [7]):

$$D_A(z) = \frac{\chi(z)}{1+z} \tag{24}$$

7. **Cluster Center Identification:** The dynamical center is approximated using either the coordinates of the Brightest Cluster Galaxy (BCG) or the spatial median of all cluster members

[2, 4]:

$$(\alpha_c, \delta_c) = (\alpha_{\mathrm{BCG}}, \delta_{\mathrm{BCG}}) \quad \text{or} \quad (\mathrm{median}(\alpha), \mathrm{median}(\delta))$$

8. **Angular Separation (θ):** Applying the spherical law of cosines, we compute the angular distance between the i-th galaxy and the cluster center [5]. Note that right ascension (α) and declination (δ) must be converted to radians:

$$\theta_i = \arccos\left(\sin\delta_c \sin\delta_i + \cos\delta_c \cos\delta_i \cos(\alpha_c - \alpha_i)\right) \tag{25}$$

9. **Projected Physical Radius:** The angular separation is converted into a transverse physical distance using the angular diameter distance:

$$r_{\mathrm{proj},i} = \theta_i \, D_{A,\mathrm{kpc}} \tag{26}$$

10. **Critical Density:** The density required for a spatially flat Universe:

$$\rho_{\mathrm{crit}} = \frac{3H_0^2}{8\pi G} \tag{27}$$

11. **Virial Mass Estimation (M_{200}):** Assuming the cluster is a relaxed, dynamically thermalized system, the virial theorem links the observed velocity dispersion to the total mass enclosed within the characteristic radius R_{200} [4]:

$$\sigma_{\mathrm{obs}}^2 = \frac{GM_{200}}{3R_{200}}$$

$$M_{200} = \frac{4}{3}\pi \, 200 \, \rho_{\mathrm{crit}} \, R_{200}^3 \tag{28}$$

Luminous Mass

12. **Luminosity Distance ($z \ll 1$):** For local clusters, we can approximate the luminosity distance using the linear Hubble law evaluated at the median redshift of the cluster [5]:

$$D_L \approx \frac{c\, z_{\mathrm{cluster}}}{H_0}, \qquad z_{\mathrm{cluster}} = \mathrm{median}(z_i) \tag{29}$$

13. **Distance Modulus (μ):** The difference between apparent and absolute magnitude, where D_L is expressed in parsecs (Eq. 25 [7]):

$$\mu = 5\log_{10}(D_L) - 5 \tag{30}$$

14. **Absolute Magnitude (M_r):** Derived from the apparent magnitude in the r-band (m_r),

corrected for the distance modulus and the line-of-sight Galactic extinction (A_r) [5]:

$$M_r = m_r - A_r - \left[5\log_{10}\left(\frac{D_L}{10\,\text{pc}}\right)\right] \tag{31}$$

15. **Galaxy Luminosity (L_r):** Converted from absolute magnitude and expressed in units of Solar Luminosity ($L_\odot$):

$$L_r = 10^{0.4(M_{r\odot} - M_r)} \tag{32}$$

16. **Stellar Mass per Galaxy ($M_{\text{star},i}$):** The stellar mass is estimated by applying a theoretical Mass-to-Light ratio ($\Upsilon_r = M/L \approx 2$) typical for early-type galaxies in clusters [3]:

$$M_{\text{star},i} = \Upsilon_r\, L_r \tag{33}$$

17. **Total Stellar Mass:** The sum of the individual stellar masses of all observed cluster members:

$$M_{\text{star}} = \sum_{i=1}^{N} M_{\text{star},i} \tag{34}$$

18. **Intracluster Gas Mass:** The X-ray emitting hot gas constitutes a major fraction of the cluster's baryons. This is estimated using empirical scaling relations from observations (Section 3 [6]) 1.2:

$$M_{\text{gas}} = M_{500} \cdot 0.093 \left(\frac{M_{500}}{2 \cdot 10^{14} M_\odot}\right)^{0.21} \tag{35}$$

(where $M_{500} \approx 0.7 \cdot M_{200}$ 1.2 and $h = 0.7$)

19. **Total Baryonic (Luminous) Mass:** The sum of the stellar component and the intracluster medium (gas):

$$M_{\text{lum}} = M_{\text{star}} + M_{\text{gas}} \tag{36}$$

20. **Characteristic Baryonic Radius (r_{bar}):** Defined analogously to R_{200}:

$$r_{\text{bar}} = \left(\frac{3M_{\text{lum}}}{4\pi \cdot 200\, \rho_{\text{crit}}}\right)^{1/3} \tag{37}$$

21. **Baryonic Velocity Dispersion:** The theoretical velocity dispersion predicted if the cluster's dynamics were governed solely by visible baryonic matter:

$$\sigma_{\text{bar}} = \sqrt{\frac{GM_{\text{lum}}}{3r_{\text{bar}}}} \tag{38}$$

Dark Matter Evidence and Simulation

22. **Dark Matter Mass Subtraction:** Because the dynamical mass (M_{200}) derived from the Virial Theorem is significantly larger than the luminous mass, the Dark Matter content is

found simply by subtracting the baryonic mass:

$$M_{\rm DM} = M_{200} - M_{\rm lum} \tag{39}$$

23. **Total Simulated Mass:** To dynamically simulate the cluster using an interactive mass fraction slider (f):

$$M_{\rm tot,sim} = M_{\rm lum} + f\, M_{\rm DM} \tag{40}$$

24. **Total Simulated Velocity Dispersion:**

$$\sigma_{\rm sim} = \sqrt{\frac{GM_{\rm tot,sim}}{3R_{200}}} \tag{41}$$

25. **Observed Histogram:**

$$H_{\rm obs} = \mathrm{hist}(v_{\rm obs})$$

26. **Generate Simulated Velocities:** Extract a normal probability distribution representing the galaxy kinematics, centered on the mean cluster velocity:

$$v_{\rm sim} \sim \mathcal{N}\left(\mu = \bar{v},\ \sigma = \sigma_{\rm sim}\right)$$

27. **Simulated Histogram:**

$$H_{\rm sim} = \mathrm{hist}(v_{\rm sim})$$

Data visualization

- Plot histograms: number of galaxies on Y-axis, velocity on X-axis.
- Curves: observations (blue), simulated (green), baryonic (red).

Cluster Mass via Virial Theorem

1. **Angular Separation ($\Delta\theta$)** from right ascension (α) and declination (δ) data (converted to radians, where c denotes the cluster center and i the individual galaxy):

$$\Delta\theta = \arccos\left[\sin(\delta_c)\sin(\delta_i) + \cos(\delta_c)\cos(\delta_i)\cos(\alpha_c - \alpha_i)\right] \tag{42}$$

2. **Projected Radius (R_P)** from angular separation, assuming a distance $D = 100$ Mpc (radius converted to kpc):

$$R_P = D \cdot \Delta\theta \tag{43}$$

3. **Observed Velocity ($V_{\rm obs}$)** from redshift data (or radial velocity RV):

$$V_{\rm obs} = c \cdot z \quad \text{or} \quad V_{\rm obs} = \mathrm{RV} \tag{44}$$

4. **1-D Velocity Dispersion ($\sigma_{\mathbf{1D}}$)** (observation):

$$\sigma_{1\mathrm{D}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(V_i - \bar{V})^2} \tag{45}$$

5. **3-D Velocity Dispersion ($\sigma_{\mathbf{3D}}$)** (assuming isotropy):

$$\sigma_{3\mathrm{D}} = 3\,\sigma_{1\mathrm{D}} \tag{46}$$

Baryonic components

6. **Convert Distance D (in Mpc) to pc** and compute the **Distance Modulus (μ)**:

$$\mu = 5\log_{10}\left(\frac{D_{\mathrm{pc}}}{10\ \mathrm{pc}}\right) \tag{47}$$

7. **Absolute Magnitude (M)** using apparent magnitude ($m = \mathrm{bmag}$):

$$M = m - \mu \tag{48}$$

8. **Luminosity (L)** of each galaxy from absolute magnitude (M):

$$L = 10^{-0.4(M - M_{\odot})} \tag{49}$$

9. **Total Luminosity of the cluster ($L_{\mathbf{total}}$):**

$$L_{\mathrm{total}} = \sum_{i=1}^{N} L_i \tag{50}$$

10. **Stellar Mass ($M_{\mathbf{lum}}$)** from luminosity (Υ is the mass-to-light ratio):

$$M_{\mathrm{lum}} = \Upsilon \cdot L_{\mathrm{total}} \tag{51}$$

11. **Gas mass:**

$$M_{\mathrm{gas}} = M_{500} \cdot 0.093\left(\frac{M_{500}}{2 \cdot 10^{14} M_{\odot}}\right)^{0.21} \tag{52}$$

(where $M_{500} \approx 0.7 \cdot M_{200}$ and $h = 0.7$) 1.2 1.2

12. **Total Baryonic Mass:**

$$M_{\mathrm{lum}} = M_{\mathrm{star}} + M_{\mathrm{gas}} \tag{53}$$

Notes:

The formula used to estimate the intracluster gas mass derived from Equation 12 in Giodini et al. (2009) [6].

Step 1: Definition of the gas mass fraction

By definition, the gas mass fraction within the radius R_{500} is the ratio of the gas mass to the total mass enclosed within that same radius:

$$f_{500}^{\text{gas}} = \frac{M_{\text{gas}}}{M_{500}} \tag{54}$$

From this definition, the gas mass can be isolated as:

$$M_{\text{gas}} = M_{500} \cdot f_{500}^{\text{gas}} \tag{55}$$

Step 2: Giodini's empirical relation (Eq. 12)[6]

An empirical scaling relation to calculate the gas fraction f_{500}^{gas} as a function of the total mass M_{500}:

$$f_{500}^{\text{gas}} \left(\frac{h}{0.7}\right)^{3/2} = (9.3 \times 10^{-2}) \left(\frac{M_{500}}{2 \times 10^{14} M_{\odot}}\right)^{0.21} \tag{56}$$

Step 3: Applying the Hubble parameter ($h = 0.7$)

The dimensionless Hubble parameter is set to $h = 0.7$. Substituting this value into the left side of Giodini's equation simplifies the scaling factor:

$$\left(\frac{0.7}{0.7}\right)^{3/2} = 1^{3/2} = 1 \tag{57}$$

Consequently, the equation for the gas mass fraction reduces to:

$$f_{500}^{\text{gas}} = 0.093 \left(\frac{M_{500}}{2 \times 10^{14} M_{\odot}}\right)^{0.21} \tag{58}$$

Step 4: Gas mass

Substituting the simplified gas fraction expression (Eq. 58) back into the initial mass definition (Eq. 55) yields the structure of the applied formula:

$$M_{\text{gas}} = M_{500} \cdot 0.093 \left(\frac{M_{500}}{2 \cdot 10^{14} M_{\odot}}\right)^{0.21} \tag{59}$$

Notes:

The conversion factor between M_{500} and M_{200} is a standard and robust approximation in cosmology.

Rather than calculating it via the full mathematical derivation of the NFW density profile, observational and theoretical models demonstrate that the relationship holds primarily due to the mass distribution geometry of standard clusters.

The mass enclosed within the overdensity radius R_{500} is consistently approximated to be around **70% to 72%** of the total virial mass enclosed within R_{200}. Thus, the conversion is applied directly in the calculations as:

$$M_{500} \approx 0.7 \cdot M_{200} \tag{60}$$

References in the Literature:

- **White (2001) [15]:** This paper provides the analytical background for comparing mass definitions. In Figure 1, the curve corresponding to M_{500} is a horizontal line constant at ≈ 0.72 independent of Ω_m.
- **Hu & Kravtsov (2003) [8]:** Detail the analytical transformation between different overdensity mass definitions, providing the generalized geometric mass ratio required for the M_{500} / M_{200} conversion.

Virial Theorem Derivation

11. **Gravitational Potential Energy** (light particle m orbiting heavy particle M):

$$U = -\frac{GMm}{r} \tag{61}$$

12. **Gravitational Force**:

$$F_G = \frac{GMm}{r^2} \tag{62}$$

13. **Centripetal Force**:

$$F_c = \frac{mv^2}{r} \tag{63}$$

14. **Equating Forces** ($F_G = F_c$) to find v^2:

$$v^2 = \frac{GM}{r} \tag{64}$$

15. **Kinetic Energy** (K, neglecting heavy particle's kinetic energy):

$$K = \frac{1}{2}mv^2 \tag{65}$$

16. **Replacing v^2 in K**:

$$K = \frac{1}{2}m\left(\frac{GM}{r}\right) \tag{66}$$

17. **Replacing Potential Energy U in K** :

$$K = -\frac{1}{2}U \tag{67}$$

18. **Virial Theorem**:

$$2 < K > + < U >= 0 \tag{68}$$

Cluster Virial Mass and Density

- **Context:** In a cluster, the total potential energy is the sum of all pairwise interactions, and the total kinetic energy is the sum of the kinetic energies of all galaxies. The Virial Theorem relates these totals. For an isotropic and stable system, the velocity is replaced by the velocity dispersion (σ).

19. **Total Virial Mass of the cluster (M_{Virial})** (using σ_{3D} and characteristic radius R):

$$M_{\text{Virial}} = \frac{3\,\sigma_{1D}^2\,R}{G} \tag{69}$$

20. **Total Volume** (approximated as a sphere):

$$\text{Volume} = \frac{4}{3}\pi R^3 \tag{70}$$

21. **Luminous Density (ρ_{lum})**:

$$\rho_{\text{lum}} = \frac{M_{\text{lum}}}{\text{Volume}} \tag{71}$$

22. **Total Density (ρ_{total})**:

$$\rho_{\text{total}} = \frac{M_{\text{total}}}{\text{Volume}} \tag{72}$$

Data visualization

- **X-Axis:** Radius (R).
- **Y-Axis (Quantities to Plot):**
  1. ρ_{lum} (Luminous Density, Red)
  2. ρ_{DM} (Dark Matter Density, Green)
  3. ρ_{total} (Total Density, Blue)
  4. M_{lum} (Luminous Mass, Red)
  5. M_{DM} (Dark Matter Mass, Green)
  6. M_{total} (Total Mass, Blue)
  7. V_{obs} (Observed Velocities, Blue data points)
  8. V_{sim} (Simulated Velocities, Green)
  9. V_{bar} (Baryonic Component Velocities, Red)

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